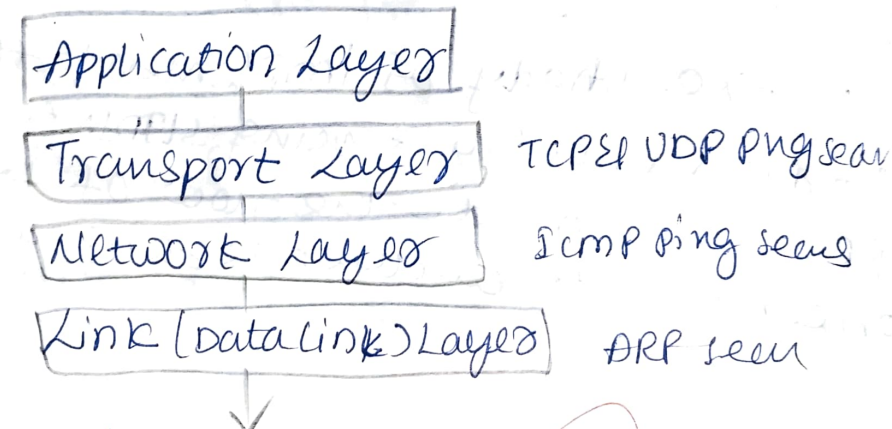
Lab Experiment-8

AIM:-

To discover the hosts on a network using Nmap post-discovery scans ARP scan, ICMP scan, and TCP/UDP ping scan - and to understand which protocol operates at which layer of the TCP/IP model, and when each one applies. Nmap can determine whether a host is alive using four different protocols.

* ARP
* ICMP
* TCP
* UPP

TCP/IP Layers and the protocol Nmap use four host Discovery



Steps to Lab check List:-

1. Log into Try HackMe and open "Nmap Live host Discovery" room.

2. work through Tasks 1-3 and answer the Simulator based questions

3. Replay - use tap network in task 4 and start the attackBox, note the target subnet 1 CONNECTION - IP. (D) Run an ARP scan, sudo nmap -PR -sn <subnet>/24 and record how many hosts respond.

5. Run an icmp echo scan; sudo nmap -PE -sn <subnet>/24 - compare the result with the ARP scan

6. Run an icmp Timestamp(-PP) and icmp address Mask (-PM) scans; note which ones are filtered

7. Run a TCP SYN Ping (-PS), a TCP ACK Ping (-PA, needs sudo) and UDP ping (-PU); compare results and required privilege level for each

8. Optionally capture traffic with hp dump or wireshark on the tun0 interface while each scan runs to see the raw ARP/icmp/TCP/UDP packets.

9. Try nmap -sL <subnet>/24 and nmap <subnet>/24 to see target listing and reverse-DNS behaviour

10. complete the rooms knowledge - check questions in Tasks 5-9, then move on to the "Nmap Basic Port Scans" room.

Result:-

Thus the steady to discover live hosts on

Lab Experiment-7

Implementation of subnetting using cisco packet tracker.

AIM:-
To implement class less IP subnetting using cisco packet tracer

Steps:-
* open cisco packet area
* create a new network
* Add routers, switches and PCs.
* connect devices using suitable cables.

Subnetting:-

192.168.1.0/24 using 27 gives: 8 subnet
30 usable host addresses per subnet.

configure router:-

open Router → CLI
enable
configure terminal
interface Fastethernet 0/0
ip address 192.168.1.1 255.255.255.224
no shutdown
exit.
for another interface
interface fastethernet 0/1
ip address 192.168.2.1 255.255.255.224

no shutdown
exit
gigabit ethernet
enable
configure terminal
interface gigabit ethernet 0/0
ip address 192.168.1.1 255.255.255.224
no shutdown
exit

interface fastethernet 0/2
switchport mode access
exit

configure PC's

open:

PC → config → IP configuration

Router:
IP address
Subnet mask: 255.255.255.224
Default gateway: Routers IP address

Test of network
1) Open PC → command prompt
2) using ping cmd
3) Ping another PC to router
Ex: Ping 192.168.1.1
If replies are received, network is working correctly.

Result:
Thus, classless IP subneting was successfully